Ghosts, Geometry, and Reality in Fourth-Order Gravity

A thematic guide to the manuscript programme

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Abstract

This document is a guide, not another synthesis. It explains how the papers in the programme fit together, which question each paper is designed to answer, which conclusions may be carried forward, and which stronger interpretations are explicitly excluded.

The manuscript series follows four connected threads. Papers 01–06 study positive and indefinite completions of fourth-order systems and the interaction problem. Papers 07–13 move to gauge reduction, causal BV–BFV transport, compact laboratories, observables, anomaly, and nonlinear continuation. Papers 14–18 study pure-Weyl black holes and the four-level ghost classification. Papers 19–22 separate physical assumptions from mathematical carriers through Mannheim gravity, gauge algebra, reverse foundations, and a Bateman–Turok counterfamily. Papers 98 and 99 provide physicist and general-audience entrances; Papers 90–92 and the computational supplements expose selected interfaces and verification details.

The guide is organized by reading purpose rather than release order. A reader can enter through a physical question—ghosts, causal gauge theory, black holes, galactic data, or mathematical foundations—and follow only the papers and evidence rails needed for that question.

Authorship and use

Asger Alstrup Palm, a computer scientist and Honorary Professor at DTU Compute, orchestrates the programme and is the accountable human contact. AI systems perform substantial derivation, programming, literature work, adversarial review, and drafting. Claude Fable 5 contributed to this guide. Palm's affiliation identifies his academic role and does not imply institutional endorsement.

The papers are research manuscripts, not substitutes for independent expert review. This guide describes their intended logical relationships; it does not strengthen any theorem. Follow each technical claim to its assumptions, certificate, verifier, and explicit non-claims in the public repository.

Contents

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1 Choose an entrance

The three public-facing documents have deliberately different jobs.

Reader	Begin with	Purpose
General audience	**Paper 99**	Why a theory depends on physical, mathematical, logical, operational, and empirical assumptions.
Physicist or referee	**Paper 98**	Compact technical synthesis, exact claim boundaries, and routes into the certificate atlas.
Reader choosing a specialist route	**Paper 00** (this guide)	Which manuscript answers which question, how papers depend on one another, and where an inference must stop.

For a single theme, the shortest routes are:

- **What is a ghost?** Read Papers 01, 03, 04, 05, and 15.
- **What survives gauge reduction?** Read Papers 07, 08, 10, and 15.
- **What survives interaction?** Read Papers 05, 06, 11, and 13.
- **What is the quantum obstruction?** Read Paper 12, then Paper 98 for the Lorentzian boundary.
- **What happens around Schwarzschild?** Read Papers 16 and 17; use Papers 14 and 18 for architecture and the separate static problem.
- **What must Mannheim gravity assume?** Read Papers 19 and 18.
- **What changes when the mathematics changes?** Read Papers 20, 21, and 22.

2 How claims move through the series

The series began from a four-level distinction that remains the safest way to read it. The word "ghost" can denote:

1. a local solution of a differential equation;
2. a classical direction surviving gauge symmetry, constraints, charges, boundaries, and the symplectic radical;
3. a direction admitting consistent nonlinear continuation;
4. a state or particle in a positive quantum theory.

A paper can establish one level while leaving the next untouched. This is why the guide gives every manuscript both a *use* and a *boundary*. The boundary is part of the result, not a disclaimer added afterward.

Quantum claims also retain one or more exact dependency tags:

`LOCAL-ALGEBRAIC  EUCLIDEAN-SPECTRAL  REDUCED-MODE  LORENTZIAN-CAUSAL`.

A Euclidean determinant does not become a Lorentzian state. A reduced mode does not become a full metric/BV propagator. A residual cohomology class does not become a one-particle graviton without a separate transfer theorem.

How to use the guide. Choose a question, read the paper listed for that question, and stop at the boundary stated in the same row. Cross the boundary only through an explicit paper, bridge note, or certificate.

3 Thread I: completion and interaction—Papers 01–06

This thread starts with the Pais–Uhlenbeck oscillator, separates positive metrics from vacuum and representation data, carries the distinction into gravity, and then tests whether free completions survive interactions.

Paper 01: canonical positive symplectic diagonalization. Use this paper for its exact unequal-frequency normal form and positive finite-dimensional metric. Do not infer a field representation, equal-frequency completion, interacting metric, or gravitational state.

Paper 02: minimum distortion and field representation. Use this paper to distinguish the variationally selected oscillator metric from the ultraviolet requirements of a fourth-order field representation. Do not treat a finite similarity transform as an automatically implementable Fock-space transformation.

Paper 03: vacuum, Fock sectors, and the Krein boundary. Use this paper for the classification of vacuum covariance, metric orbits, Fock sectors, and the equal-frequency Jordan limit. Do not infer that a Krein carrier selects a positive physical state or survives interaction.

Paper 04: gauge reduction in fourth-order gravity. Use this paper to see how Lorentz covariance, helicity, constraints, and the conformal Jordan boundary alter the scalar completion problem. Do not model the gravity system as independent copies of the oscillator or infer a nonlinear quantum theory.

Paper 05: resonant interaction obstruction. Use this paper for the scoped failure of the canonical analytic positive metric under branch-changing interaction in fourth-order scalar systems. It does not exclude indefinite, nonlocal, singular, or nonperturbative prescriptions.

Paper 06: Einstein–Weyl interaction obstruction. Use this paper for cubic protection followed by a second-order metric failure in a declared gravitational channel. Do not promote the obstruction to all interactions, all backgrounds, or strict pure-Weyl gravity.

Thread logic: Paper 01 constructs a positive finite model; Papers 02–04 expose the extra representation and gauge obligations; Papers 05–06 test the first nonlinear promotion. The correct conclusion is not "positive" or "ghostly" in the abstract, but a typed map of which completion survives which gate.

4 Thread II: causal and compact pure-Weyl theory—Papers 07–13

This thread replaces local mode counting with full complexes, support conditions, residual reduction, action-derived pairings, relational observables, anomaly classes, and second-order continuation.

Paper 07: residual cohomology on the conformal cylinder. Use this paper for the declared zero-charge residual reduction and the centered Weyl-square classes. The classes are deformation or vertex classes, not one-particle gravitons, and their positive residual pairing is not a positive interacting Hilbert space.

Paper 08: covariant causal transport. Use this paper for the complete free BV–BFV transport between support conditions and for the transported pairing. Do not infer interacting causal products, a positive Hadamard state, or a Lorentzian QME.

Paper 09: a backreacting phase clock. Use this paper for a controlled compact relational-clock construction and its Cartan analysis. The fixture is a model laboratory, not a realistic matter clock, cosmology, or universal resolution of the problem of time.

Paper 10: Einstein–Maxwell inside Weyl–Maxwell. Use this paper for the exact inclusion of the Einstein–Maxwell tangent and the identification of additional nonradical branches on a compact product. Classical quotient directions and Lee–Wald signs are not quantum particles or probabilities.

Paper 11: retained gravity–light bracket. Use this paper for an exact mixed bracket surviving cyclic homological reduction through the declared third operation. Do not infer an all-order interacting observable algebra or scattering construction.

Paper 12: local one-loop BV obstruction. Use this paper for the strict local Euclidean anomaly classification and the formal compensator restoration in an enlarged theory. The compensator changes the field content; neither branch supplies a Lorentzian QME, positive state, particles, or unitarity.

Paper 13: formal second-order tangent cone. Use this paper for stabilizer moment maps, finite-harmonic continuation, and declared bounded-resonance tests. A formal second-order jet is not an exact family, an all-orders solution, or a complete bounded nonlinear cone.

Thread logic: Papers 07–08 construct the free residual and causal base. Papers 09–11 ask how clocks, phase space, and observables sit on that kind of structure. Papers 12–13 test two different nonlinear promotions: quantum consistency and classical second-order continuation. Their outcomes do not substitute for one another.

5 Thread III: black holes and the four-level classification—Papers 14–18

The black-hole thread separates operator factorization, endpoint selection, resonance, causal response, and static charges. It should not be read as one undifferentiated ringdown claim.

Paper 14: black-hole radiation architecture. Use this paper for the Ricci–Bach composition, horizon pairing, and the operator architecture that motivates the later endpoint and resonance theorems. For theorem-strength endpoint and double-pole claims, continue to Papers 16 and 17 rather than treating the architecture as their substitute.

Paper 15: four-level ghost classification. Use this paper to compare local solutions, classical reduced directions, nonlinear continuation, and positive quantum states across the controlled laboratories. It is a synthesis of scoped results, not a universal rescue or universal no-go theorem.

Paper 16: Lorentzian endpoint non-selection. Use this paper for the theorem that future-horizon regularity and ordinary one-sided radiative traces do not select the Einstein subsector of the complete axial system in the stated scope. Do not infer all-multipole, polar, nonlinear, or quantum non-selection.

Paper 17: non-split extension and defective resonance. Use this paper for the repeated Regge–Wheeler self-extension, one certified Smith type $(0, 0, 2)$, the nonzero double radial Green pole, and its retarded mode-reduced parent. It does not establish a full metric/BV causal resolvent, global contour theorem, complete late-time expansion, real astrophysical excitation, or detector waveform.

Paper 18: static Bach-flat charges and first laws. Use this paper for residual-basic normalization of Hamiltonian, entropy, and signed temperature across the certified static Mannheim–Kazanas family. It is a static charge theorem, not a physical-process or radiative thermodynamics theorem.

Thread logic: Paper 14 supplies architecture; Paper 15 supplies the cross-laboratory claim grammar; Paper 16 answers the endpoint-selection question; Paper 17 answers a spectral and mode-reduced causal-response question; Paper 18 deliberately returns to the separate static problem.

6 Thread IV: assumptions and reverse foundations—Papers 19–22

This thread asks which parts of a physical conclusion come from the equation, which come from matter and boundary premises, which come from the gauge or mathematical carrier, and which survive explicit counterexamples.

Paper 19: what conformal gravity must assume. Use this paper to separate the Mannheim–Kazanas radial solution from its matter, conformal-frame, source-moment, boundary, and galactic-data premises. Its NGC 3198 comparison is one frozen 39-point protocol, not population-level model selection or a universal verdict on conformal gravity.

Paper 20: direct check of the Bach gauge algebra. Use this paper for a polynomial-jet search for Noether identities. Within the tested derivative orders and curvature degrees, it finds no additional local generator after quotienting differential reparametrizations and trivial symmetries. It is a lower bound on completeness, not an all-orders closure theorem.

Paper 21: reverse foundations of physics. Use this paper for the literature synthesis and typed separation of logic, existence theory, mathematical carrier, physical postulates, and one physical obligation. Its $6 \times 6 \times 16 = 576$ atlas coordinates are questions, not complete theories; populated cells and programme coverage do not compose without interface theorems.

Paper 22: a Bateman–Turok Green-tail counterfamily. Use this paper for the exact positive, nonseparable torus family that refutes one deterministic all-field scaled-PL route. It does not decide Gibbs typicality, the interacting Witten or H^{-1} estimate, continuum reconstruction, positivity, or any Lorentzian claim.

Thread logic: Paper 19 separates a solution from its physical interpretation. Paper 20 tests whether a familiar algebra is premise or consequence. Paper 21 turns that separation into a general research atlas. Paper 22 demonstrates how the atlas should behave when an attractive bridge fails: refute the bridge precisely without inflating the result into a verdict on the entire programme.

7 Public syntheses, supplements, and bridge notes

7.1 Public syntheses

Paper 98: Reverse physics and pure-Weyl gravity. The physicist-facing synthesis. Use it for the compact programme argument, exact dependency boundaries, principal numerical comparisons, and reviewer route into the interactive evidence atlas.

Paper 99: How to Build a Universe. The general-audience introduction. Use it for the theory-as-a-stack idea, the purpose of the 576-coordinate atlas, and the role of exposition and verification in AI-assisted research.

7.2 Computational supplements and archival manuscript

The supplements document derivations, conventions, tables, and reproduction commands without changing the claim boundary of their parent paper:

- Papers 07–08 computational supplement;
- Paper 09 computational supplement;
- Paper 12 computational supplement;
- Paper 16 computational supplement.

The combined Papers 07–08 archival manuscript preserves the earlier paired presentation; use the separate Papers 07 and 08 for the cleaner residual/causal division.

7.3 Bridge notes 90–92

Bridge notes translate between communities and expose interfaces; they are not additional headline-paper conclusions.

Paper 90: cyclic Green transfer. Connects Green-hyperbolic complexes, support-local cyclic deformation retracts, and conformal detour constructions. Use it with Papers 08 and 11.

Paper 91: charge-fibre Taub bridge. Connects compact pure-extra obstruction and a balanced Einstein–extra second-order extension. Use it with Papers 10 and 13; its formal second-order jet is not an all-orders family.

Paper 92: compact currents and black-hole endpoints. Compares extra classical Lee–Wald directions with adjacent positive-metric, BRST/Fock, and boundary-selection questions. Use it between Papers 10, 16, and 17; classical signs remain distinct from quantum norms.

8 Reading routes by research question

Question	Route
Can a fourth-order free theory have a positive completion?	01 → 02 → 03 → 04.
Does that completion survive interaction?	05 → 06; compare 13 for a different nonlinear gate.
What survives the classical gauge quotient?	07 → 08 → 10; synthesize with 15.
Can an observable or clock be transported?	09 and 11, with Bridge 90 for the causal-transfer interface.
Where does quantization first fail or remain incomplete?	12 → 98; keep Euclidean anomaly and Lorentzian pseudo-state as separate questions.
Do black-hole boundary conditions remove the extra sector?	14 → 16 → 17; use 18 only for the separate static charge problem.
Does conformal gravity explain rotation curves?	19, then 18 for static geometry and 21 for the assumption map.
Is the gauge algebra assumed or derived?	20 → 21.
What if Choice, Hilbert space, or the continuum is changed?	21, using 03 for Krein structure and 22 for a concrete Euclidean counterexample.
How should an external reviewer begin?	98 → relevant technical paper → certificate → independent verifier.

9 How to inspect a claim

The Reverse Physics Atlas is the interactive companion to this guide. It exposes the dimensions, matrix cells, theory profiles, programme assemblies, end-to-end journeys, Weyl BV routes, literature records, certificates, and formal-proof passports.

For a computer-assisted result, follow

$$\begin{aligned}\text{paper claim} &\longrightarrow \text{producer} \longrightarrow \text{machine-readable certificate}\\ &\longrightarrow \text{independent verifier} \longrightarrow \text{adversarial or mutation test.}\end{aligned}$$

Re-running the producer is reproduction, not independent verification. Exact algebra and numerical evidence are different types. A skip, missing input, or timeout is not a pass.

The manuscript sequence alone is not a dependency proof. Two papers may use different actions, backgrounds, carriers, support conditions, or empirical protocols. A later paper can sharpen an earlier question without promoting all of its claims. The certificate and non-claim boundary determine what may be transferred.

10 What the whole series does and does not claim

Across the four threads, the programme establishes examples of positive and Krein free completions, interaction obstructions, residual and causal gauge constructions, additional nonradical classical directions, nonlinear balance and obstruction, a strict local Euclidean anomaly, a full-complex indefinite BRST–Hadamard pseudo-state pair, black-hole endpoint non-selection and a defective mode-reduced response, bounded empirical controls, and a deterministic Euclidean counterfamily.

Those ingredients do not compose automatically into a complete theory. The series does not establish a positive full-BV particle space, Lorentzian interacting quantum gravity, scattering or unitarity, a complete causal black-hole waveform, population-level galactic model selection, a continuum Bateman–Turok reconstruction, or a universal weakest-foundation theorem.

Role of Paper 00. Paper 99 explains why the programme matters. Paper 98 states the compact physicist-level synthesis. Paper 00 tells the reader where to go next and where each route must stop.